



TCM-Sage

A Glass-Box Evidence-Synthesis System for Classical Traditional Chinese Medicine

Hybrid RAG + Knowledge Graph Integration for Traceable Classical TCM Evidence Retrieval

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github.com/AndyZHENG0715/TCM-Sage



1 Background & Motivation

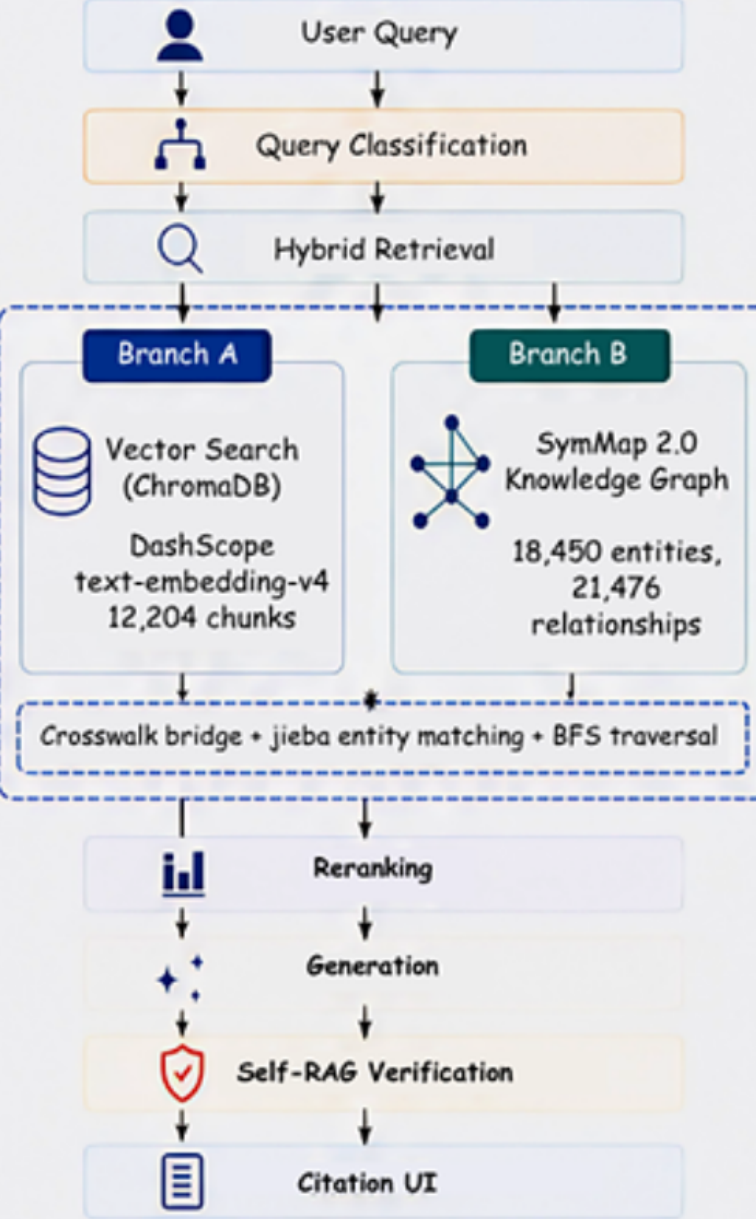
- TCM literature spans thousand of years and is vast and fragmented.
- Manual search is slow and cross-text comparison is inefficient
- Black-box LLMs lack traceability and verifiability.
- We build a transparent assistant that retrieves, synthesizes, and verifies classical TCM evidence with citations.

Goal: Build an evidence-synthesis assistant that helps practitioners retrieve, synthesize, and verify classical TCM evidence with direct source citations.

2 Objectives

- Build a modular RAG system for classical Chinese medicine texts.
- Integrate a knowledge graph to bridge modern and classical terminology.
- Provide passage-level citation tracing through a glass-box interface.
- Validate usefulness with blind A/B evaluation and statistical testing.

3 System Overview / Methodology



4 Knowledge Base & Retrieval Design

17 classical TCM texts	17
3.72M characters	3.72M
12,204 indexed chunks	12,204
Clause-level chunking for Shanghan Lun (388 clauses)	388
Clause-level chunking for Jingui Yaolue (489 clauses)	489

Most texts are chunked by chapters to preserve context and improve efficiency. Key clinical classics (Shanghan Lun and Jingui Yaolue) are chunked by clauses to enable precise retrieval of formulas and clauses.

- Formula-aware canonical retrieval**
Map variant names to canonical formulas and retrieve with domain-aware normalization
- Source authority boost**
Weight passages by text authority, time period, and clinical relevance.
- qwen3-rerank**
High-quality reranking using qwen3-rerank for better relevance and precision

5 Key Features

1 Evidence-Based Answers

You
What are the indications of Guizhi Tang?

TCM-Sage
Guizhi Tang is commonly used for exterior wind-cold patterns with aversion to cold, mild fever, headache, and a floating pulse

Citations
Shanghan Lun · Chapter 2 · 1

Verified

2 Citation Panel

Classic Text Source
Shanghan Lun · Chapter 2

Passage Content
"Taiyang disease with aversion to cold, no sweating, headache, body aches, and a floating pulse: this is called wind pattern."

View Full Passage

Relevance: 95.0%

3 KG Explorer



4 Arena Blind A/B Evaluation

Blind comparison (identity hidden)

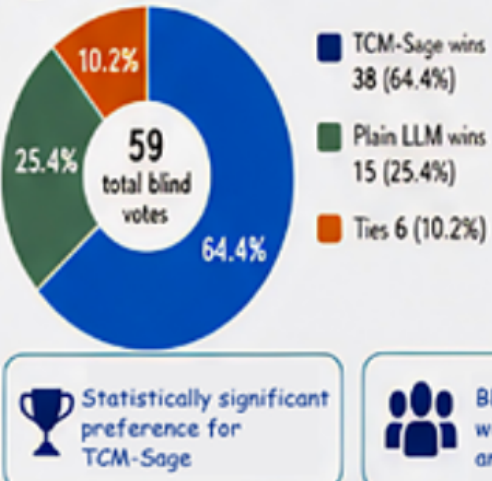
Model A	Model B

A is better Tie B is better

5 Multi-Provider LLM Support

- Qwen
- OpenAI / GPT
- Claude
- Gemini
- Local LLMs

6 Evaluation Results



7 Contributions

- Hybrid RAG pipeline for classical Chinese medicine
- SymMap 2.0 integration with a crosswalk bridge for ancient-modern terminology
- Domain-specific retrieval optimization: clause-level chunking, formula-aware retrieval, and authority boosting
- Glass-box UI with traceable citations and verification.
- Arena evaluation framework with statistical validation.

8 Conclusion

- TCM-Sage provides transparent, evidence-based answers with full citations.
- Hybrid retrieval and domain-aware optimization improve relevance and precision.
- User evaluation shows a significant preference over a plain LLM.
- Supports safe, traceable AI assistance for learners and practitioners.

9 Future Work

- Corpus Expansion**
Add more texts and standardized annotations.
- Institution-Specific Knowledge Bases**
Support tailored standards and collaborative sharing.
- Patient-Friendly Explanation Mode**
Adapt explanations for different users, such as patients.
- Stronger/Fine-Tuned TCM Base Models**
Explore stronger training for better domain-aligned performance.